International Journal of Infectious Diseases 163 (2026) 108224

Contents lists available at ScienceDirect

International Journal of Infectious Diseases

journal homepage: www.elsevier.com/locate/ijid

Case Report

First case of Mpox clade Ib transmitted from human-to-human outside the African continent and not linked to travel

Jorge-Alfredo Pérez-García[1,2,*], Laura Dans-Vilan[2,3], Laura Montero-Morales[4], Esther Culebras[1,2], Santiago Fernández-Castelao[2,3], Ángel Miguel-Benito[4], Teresa Puerta-López[2,5], Alberto Delgado-Iribarren García-Campero[1,2]

[1] *Centro Sanitario Sandoval/Hospital Clínico San Carlos, Clinical Microbiology Unit, Madrid, Spain*
[2] *IdISSC. Instituto de Investigación Sanitaria Hospital Clínico San Carlos, Madrid, Spain*
[3] *Centro Sanitario Sandoval/Hospital Clínico San Carlos, Internal Medicine Unit, Madrid, Spain*
[4] *Directorate General of Public Health, Regional Ministry of Health of Madrid, Madrid, Spain*
[5] *Centro Sanitario Sandoval, Infectious Diseases Unit, Madrid, Spain*

ARTICLE INFO

Article history:
Received 24 October 2025
Revised 12 November 2025
Accepted 13 November 2025

Keywords:
Monkeypox
Sexually transmitted infections
Clade Ib
Outbreak
First case

ABSTRACT

On 14 August 2024, World Health Organization declared Mpox a Public Health Emergency of International Concern due to the risk of international spread of clade Ib. We report the first documented case of human-to-human transmission outside the African endemic regions with no history of travel in the case or their contacts. Maintaining surveillance and precautionary measures is essential to prevent a potential spread in cases caused by clade Ib in non-endemic areas. Further data on transmission and virulence are still required.

© 2025 The Authors. Published by Elsevier Ltd on behalf of International Society for Infectious Diseases. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)

Introduction

Mpox is a zoonotic infectious disease caused by the Mpox virus, a member of the genus *Orthopoxvirus*, family *Poxviridae*. Human infection with Mpox was first reported in the Democratic Republic of Congo in 1970 [1]. Following the first human case, sporadic outbreaks were reported in West Africa (clade II) and East/Central Africa (Clade I). In May 2022, the first cases of the multinational outbreaks were reported in Europe [2,3], prompting the World Health Organization (WHO) to declare the epidemic as a Public Health Emergency of International Concern (PHEIC) in July 2022 [4]. As of 31 August 2025, a total of 162,785 including 424 deaths have been reported to WHO [5]. This global outbreak is caused largely by the clade IIb strain, associated with close intimate contact (including sexual activity) and most cases are diagnosed among men who have sex with men [6,7]. Sexual transmission has been suspected from the beginning of outbreaks, mainly with the appearance of cases related with severe anal pain, proctitis [8], and genital ulcers [9]. Furthermore, the presence of another concomitant sexually transmitted pathogens in patients reveals the possibility of sexual transmission [10]. Signs and symptoms could include fever, asthenia, fatigue, headache, inguinal lymphadenopathy, proctitis, exanthema, ulcers, and vesicles on the genitals and perianal region or other skin and mucous membrane locations. Differences in pathogenicity, severity, risk groups, and transmission between clades Ib, Ia, and IIb complicate outbreak management [11]. Compared to clade II, clade I exhibits higher transmissibility and virulence, leading to more severe disease manifestations and higher mortality rates [12–14]. The emergence of clade Ib, which is more transmissible than clade Ia, has worsened the outbreak situation. On 14 August 2024, WHO declared Mpox a PHEIC due to the risk of international expansion of the new clade Ib [15]. This clade started a major outbreak in 2023 in the eastern part of the Democratic Republic of Congo and is undergoing sustained human-to-human transmission in several countries, mainly affecting African continent. The first case reported outside Africa was detected in a Swedish traveler returning from an area of Central Africa affected by clade I outbreak [16]. Since then, cases have been detected in countries outside Africa, although, to date, all have been detected in patients with a recent history of travel to endemic African countries or contact with persons who had recently made such travel [5]. On 4 September 2025, WHO considered that it was no longer a PHEIC due to sustained declined in cases and deaths in the affected countries [17].

* Corresponding author.
E-mail address: jorgealfredo.perez@salud.madrid.org (J.-A. Pérez-García).

https://doi.org/10.1016/j.ijid.2025.108224
1201-9712/© 2025 The Authors. Published by Elsevier Ltd on behalf of International Society for Infectious Diseases. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)

Figure 1. Ulcerative lesion of the patient.

The aim of this communication is to present the first case of Mpox clade Ib detected in a patient with no epidemiological history of travel to an endemic African area or contact in the last month with people who had traveled to these areas and to alert about the possible expansion of this clade Ib in non-endemic countries due to human-to-human transmission.

Case description

The patient is a 49-year-old Spanish male with medical history notable for past hepatitis B virus and genital herpes simplex virus type 2. He has been vaccinated with two doses of Invanex (modified live Ankara virus-Bavarian Nordic) against Mpox 2 years prior. His only regular medication was daily HIV pre-exposure prophylaxis with tenofovir disoproxil fumarate and emtricitabine. He went to a sexually transmitted infections clinic in Madrid, Centro Sanitario Sandoval, presenting with a single genital lesion of 5 days of evolution. A rounded lesion in the balanopreputial groove of 8 mm in diameter, with well-defined edges, a fibrinous non-necrotic base and little exudate (Figure 1), not painful to palpation. He was treated with oral valacyclovir 500 mg every 12 hours under suspicion of herpes simplex virus recurrence without clear improvement. Additionally, he reported the concurrent appearance of painful and palpable left inguinal lymphadenopathy, and general malaise without fever. Oral doxycycline 100 mg every 12 hours (due to initial suspicion of *Chlamydia trachomatis* ser. lymphogranuloma venereum) and non-steroidal anti-inflammatory drugs were started. He did not travel outside of the country in the previous months and his sexual activity in the previous 21 days, included unprotected intercourse with two local men who had not traveled abroad or had sexual contact with people from the endemic regions. The encounters took place in private homes, not in saunas or sex-on-premises venues. There was no concurrent substance use (not regular chemsex user). The patient was informed of the results of the microbiological test, and hygiene and isolation measures were explained to him. As of the date of submission of this manuscript, the patient remains in good general condition, with no new lesions or significant changes in the described lesion.

Microbiological investigation

A dark-field study and serologies for syphilis were requested, which ruled out a syphilitic etiology of the ulcer. For Mpox detection, an ulcer sample was collected, and a real-time polymerase chain reaction (PCR) was performed (Monkeypox Virus Genotyping Real-Time PCR Kit, BioPerfectus), being positive for clade I. The nucleic acid extraction was carried out in Microlab STARlet-Hamilton (Seegene) and reverse transcription-PCR in CFX96 Touch Real-Time (Bio-Rad). A different real-time PCR was performed to verify the result (STANDARD M10 MPX/OPX, SD Biosensor), confirming and differentiating clade Ib. Ulcerative pathogens other than Mpox were tested by reverse transcription-PCR Seegene Allplex® Ulcer Assay (Seegene, South Korea), being negative for all of them.

Discussion

To our knowledge, this is the first report of a case of Mpox clade Ib without epidemiological history of travel to Africa in the index patient or their previous sexual contacts, reported by Spanish Public Health authorities on 10 October 2025 [18]. Subsequent cases have recently been reported in Europe and the United States with evidence of autochthonous transmission [18,19]. Before this time, cases had already been reported across several countries, both in Europe and in the rest of the world [20], but were always associated with individuals who traveled to a country with known Mpox clade Ib transmission and were likely exposed there, or with individuals who did not travel personally but had direct contact with someone who did. In Spain, there is only one reported case of clade Ib in a patient who had traveled to Tanzania, with risk exposure occurring within the previous 21 days, specifically on September 22, and who has no epidemiological link to the case presented here [5]. This finding indicates that we should maintain surveillance despite the deactivation of the PHEIC and be alert to the possible community transmission of this clade in areas where it had not previously occurred, establish appropriate protocols for the possible international expansion of the new clade Ib, and take the necessary measures to minimize both the clinical and public health consequences, especially in groups that have been shown to be particularly vulnerable in previous Mpox outbreaks since 2022, such as people living with HIV, men who have sex with men, and persons with multiple sexual partners. Public health measures should include the implementation of control measures upon case detection to interrupt transmission and prevent the emergence of secondary cases, clade determination in all confirmed cases, vaccination of individuals at higher risk of exposure to Mpox, and maintain coordination with scientific societies, civil society, and community organizations to ensure effective public health communication and response.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Ethical approval statement

The document, as an outbreak report, does not require ethical approval. All data are anonymized and presented in compliance with ethical and privacy standards. The patient has approved both the text and the image for publication.

Declaration of competing interest

The authors have no competing interests to declare.

Acknowledgments

Our thanks to the Clinical Microbiology Unit of the Hospital La Paz, Madrid, and to Centro Nacional de Microbiología, Majadahonda (Madrid), for their help confirming the microbiological results. All authors have contributed equally to the preparation of this article.

References

[1] Breman JG, Kalisa-Ruti, Steniowski MV, Zanotto E, Gromyko AI. Arita I. Human monkeypox, 1970-79. *Bull World Health Organ* 1980;**58**:165–82.
[2] Antinori A, Mazzotta V, Vita S, Carletti F, Tacconi D, Lapini LE, et al. Epidemiological, clinical and virological characteristics of four cases of monkeypox support transmission through sexual contact, Italy, May 2022. *Eurosurveillance* 2022;**27**:2200421.
[3] Angelo KM, Smith T, Camprubí-Ferrer D, Balerdi-Sarasola L, Díaz Menéndez M, Servera-Negre G, et al. Epidemiological and clinical characteristics of patients with monkeypox in the GeoSentinel Network: a cross-sectional study. *Lancet Infect Dis* 2023;**23**:196–206.
[4] World Health Organization *Second meeting of the International Health Regulations (2005) (IHR) Emergency Committee regarding the multi-country outbreak of monkeypox*. Geneva: World Health Organization; 2022.
[5] World Health Organization *Global Mpox Trends*. Geneva: World Health Organization; 2025.
[6] Orviz E, Negredo A, Ayerdi O, Vázquez A, Muñoz-Gomez A, Monzón S, et al. Monkeypox outbreak in Madrid (Spain): Clinical and virological aspects. *J Infect* 2022;**85**:412–17.
[7] Thornhill JP, Barkati S, Walmsley S, Rockstroh J, Antinori A, Harrison LB, et al. Monkeypox virus infection in humans across 16 countries — April–June 2022. *N Eng J Med* 2022;**387**:679–91.
[8] Pipitò L, Trizzino M, Ferraro D, Cascio A. Monkeypox proctitis treated with doxycycline in an HIV MSM returning to Italy from France. *Travel Med Infect Dis* 2022;**50**:102469.
[9] Mitjà O, Ogoina D, Titanji BK, Galvan C, Muyembe JJ, Marks M, et al. Monkeypox. *Lancet* 2023;**401**:60–74.
[10] Rizzo A, Pozza G, Salari F, Giacomelli A, Mileto D, Cossu MV, et al. Concomitant diagnosis of sexually transmitted infections and human monkeypox in patients attending a sexual health clinic in Milan, Italy. *J Med Virol* 2023;**95**:e28328.
[11] Africa Centers for Disease Control and Prevention *Africa CDC Epidemic Intelligence Weekly Report*. Addis Ababa: Africa Centers for Disease Control and Prevention; 2024.
[12] Tiwari A, Kalonji T, Miller T, Van Den Bossche T, Krolicka A, Muhindo–Mavoko H, et al. Emergence and global spread of Mpox clade Ib: challenges and the role of wastewater and environmental surveillance. *J Infect Dis* 2025;**231**:e825–9.
[13] Katoto PD, Muttamba W, Bahizire E, Malembaka EB, Bosa HK, Kazadi DM, et al. Shifting transmission patterns of human mpox in South Kivu, DR Congo. *Lancet Infect Dis* 2024;**24**:e354–5.
[14] Zamani S, Noroozi E, Hasanzadeh A, Zamani E, Trinidad JC, Firooz A. Mpox infection: a state-of-the-art overview of epidemiological, molecular, and clinical aspects following the 2024 public health emergency. *J Infect Public Health* 2025;**18**:102940.
[15] World Health Organization *WHO Director-General declares mpox outbreak a public health emergency of international concern*. Geneva: World Health Organization; 2024.
[16] Treutiger CJ, Filén F, Rehn M, Aarum J, Jacks A, Gisslén M, et al. First case of mpox with monkeypox virus clade Ib outside Africa in a returning traveller, Sweden, August 2024: public health measures. *Eurosurveillance* 2024;**29**:2400740.
[17] World Health Organization *WHO Director-General's opening remarks at the media briefing –5 September 2025*. Geneva: World Health Organization; 2025.
[18] Centre for Disease Control and Prevention *Detection of autochthonous transmission of monkeypox virus clade Ib in the EU/EEA*. Atlanta: Centre for Disease Control and Prevention; 2025.
[19] Centre for Disease Control and Prevention *Monkeypox in the United States and Around the World: Current Situation*. Atlanta: Centre for Disease Control and Prevention; 2025.
[20] Shah Y, Singh AK, Acharya BK, Lamichhane M, Dhami RS, Pandey K, et al. Potential risk of local transmission of Mpox in Nepal: an emerging Public Health concern. *Health Sci Rep* 2025;**8**:e71168.